## Conclusions

In this article, prediction of ionic conductivity for a highly varied dataset for Li bearing compounds ~~was explored~~ by a mixed set of ML descriptors was explored. The importance of physicochemical descriptors is obvious, and was reported for classification of conductors from the non-conductors in our previous study, though the use of these descriptors did not provide a very accurate prediction of the magnitude of ionic conductivity as per present study. On the contrary, the use of elemental embedding in open source material science literature to derive a set of 200 virtual descriptors did not yield a very promising performance either with regard to prediction of ionic conductivity. However, a mixed basket of descriptors comprising both physicochemically relevant parameters as well embedding based virtual descriptors made possible a much accurate prediction of conductivity. The use of both descriptors in tandem was found to be critical, since ~~a~~ ~~systematic~~ experiments with reduced number of physical descriptors led to deterioration of accuracy. Understandably, the mixed set of descriptors was able to encode both structure-property relationship and the latent ~~chemical~~ signature of chemical association effectively which the individual sets of descriptors could not.

One other salient observation from this study is the superior performance of Random Forest model over boosting-based models. Given the level of noise in the data, and the extent of correlated variables in the model, the RF model provided a better fit for this prediction problem, when compared with the performance by Gradient Boosting, XG Boost, ... ... (1)

For the most part in this analysis, the prediction error was found to be normally distributed, suggesting the deviations between the predicted and actual ~~reduced~~ conductivity as largely random without strong systematic bias. Slight overprediction was observed for low conductivity materials mostly due to data imbalance in that conductivity range.

The method presented here is a cost-effective way to ~~speed up the~~ explore the search of compounds that can serve as high-performing solid electrolyte in metal ion battery. The method can serve ~~as~~ in a precursor step before undertaking trial-and-error experiments or molecular dynamics simulation to save time. Further integration ~~with~~ of graph neural network with better assimilation of molecular structure, and ~~incorporation~~ consideration of experimental uncertainties ~~in the~~ associated with the data could lead to enhanced performance ~~with~~ from the scheme, outlined in this article.